Propagation of radio waves — solution

No official solution of this experimental problem has been published. This solution was written by HaslaLab in the form of the EuPhO experimental solutions of the other years: for each task the set-up, the idea, a table of sample data, a graph and the analysis. The sample data are the readings of an emitter at $868\,\mathrm{MHz}$ *(inside the stated range* $200\,\mathrm{MHz}$–$5\,\mathrm{GHz}$*) in tap water at* $20\,^\circ\mathrm{C}$*; with another frequency the numbers change, the method does not. The receiver shows whole dBm.*

General remarks

The receiver averages for 15 s, so each point needs at least one full reading period after anything is moved. Hands, the body and the cables change the readings (the waves are reflected by everything in the room), so the same posture and the same distance to the set-up must be kept during a series, and each series is taken in one go. Every reading in dB converts to power as $P = 1\,\mathrm{mW}\cdot 10^{P_{\mathrm{dBm}}/10}$; a field amplitude decaying as $\mathrm{e}^{-\gamma z}$ gives a power $P \propto \mathrm{e}^{-2\gamma z}$, a straight line of slope $-\frac{20}{\ln 10}\gamma = -8.686\,\gamma$ in dB.

Task 1. Sensitivity of receiver. (1p)

The emitter is wrapped completely in several layers of aluminium foil (or carried as far away as possible, into the corridor). The receiver then shows only its own noise: twelve consecutive readings

−99, −105, −98, −102, −98, −101, −103, −102, −103, −102, −100, −102 dBm

scatter around $-101\,\mathrm{dBm}$. The lowest measurable power is the level of this noise floor,

$$P_{\min} \approx -100\,\mathrm{dBm} = 10^{-10}\,\mathrm{mW}.$$

Any value between 10^{-11} and $10^{-9}\,\mathrm{mW}$ shows the right order of magnitude.

Task 2. Wavelength in water. (6p)

Set-up. As in Figure 2 of the problem: the emitter lies in a cradle lined with aluminium foil on the bottom of the box E, which stands on the foam blocks G; the foil closes every gap between the box and the cradle, so the waves can leave only upwards through the water. The receiver B is held at a fixed place above the box. Water is poured in with the cup K, $100\,\mathrm{mL}$ at a time.

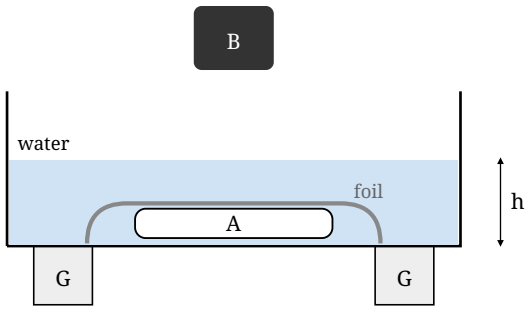


Figure S1. Set-up for Task 2 (side view).

Idea. The wave crosses the water layer of depth h and is partly reflected at the water surface (amplitude reflection $r = \frac{n-1}{n+1} \approx 0.8$ for $n \approx 9$) and again at the foil below. The layer is a Fabry–Pérot resonator: the transmitted power is largest when the round trip $2h$ holds a whole number of wavelengths in water. Successive maxima (and minima) of $P(h)$ are therefore

$$\Delta h = \lambda_{\mathrm{w}}/2$$

apart. The depth follows from the volume: the inner bottom of the box measures $29.6\,\mathrm{cm} \times 17.8\,\mathrm{cm}$, $S = 526.9\,\mathrm{cm}^2$, so $h = V/S$ and every $100\,\mathrm{mL}$ raises the water by $1.90\,\mathrm{mm}$ (checked with the ruler at the end: $h = 60.5\,\mathrm{mm}$ for $3.2\,\mathrm{L}$).

Table S1. Received power against the water depth.

V (mL)	h (mm)	P (dBm)	V (mL)	h (mm)	P (dBm)
0	0.0	−47	1700	32.3	−42
100	1.9	−47	1800	34.2	−46
200	3.8	−45	1900	36.1	−47
300	5.7	−43	2000	38.0	−48
400	7.6	−38	2100	39.9	−48
500	9.5	−30	2200	41.8	−47
600	11.4	−38	2300	43.7	−45
700	13.3	−42	2400	45.6	−42
800	15.2	−45	2500	47.4	−37
900	17.1	−47	2600	49.3	−38
1000	19.0	−47	2700	51.2	−42
1100	20.9	−47	2800	53.1	−46
1200	22.8	−46	2900	55.0	−48
1300	24.7	−44	3000	56.9	−48
1400	26.6	−40	3100	58.8	−48
1500	28.5	−34	3200	60.7	−47
1600	30.4	−38			

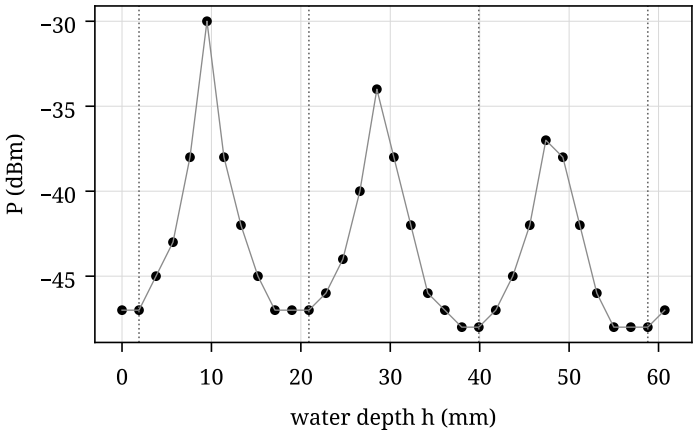


Figure S2. $P(h)$; the dotted lines mark the minima.

Analysis. The minima lie at $h = 1.9, 20.9, 39.9, 58.8\,\mathrm{mm}$ and the maxima at $h = 9.5, 28.5, 47.4\,\mathrm{mm}$. A straight-line fit of the position of the extremum against its number gives $\Delta h = 18.96 \pm 0.2\,\mathrm{mm}$ (the error is a tenth of the step in h, since the extrema are read off a curve sampled every $1.9\,\mathrm{mm}$), so

$$\lambda_{\mathrm{w}} = 2\Delta h = 37.9 \pm 0.4\,\mathrm{mm}.$$

The narrow maxima and the broad minima are what a resonator with a large reflection coefficient gives; the slow overall decrease of the curve is the attenuation in water studied in Task 3.

Task 3. Attenuation in water. (3p)

Set-up. The plastic tube D stands on its cap, taped to the bucket so that it cannot fall, filled with water and wrapped tightly with aluminium foil over its whole length: a metal waveguide filled with water. The emitter hangs on the rope M (held by the clip N at the rim) at depth z below the water surface; the receiver lies on the open end.

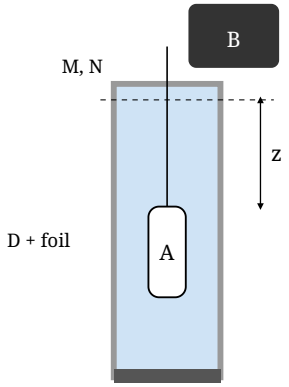


Figure S3. Set-up for Task 3.

Idea. In the water-filled guide the wave travels as in (1), $E \propto \mathrm{e}^{-\alpha_\mathrm{g} z}$, so $P_\mathrm{dBm} = \mathrm{const} - 8.686\,\alpha_\mathrm{g} z$. In water the guide is far above cut-off, so the wave is not evanescent.

Table S2. Received power against the depth of the emitter.

z (cm)	P (dBm)	z (cm)	P (dBm)
2	−17	18	−22
4	−18	20	−23
6	−19	22	−23
8	−20	24	−24
10	−20	26	−25
12	−21	28	−25
14	−21	30	−26
16	−22		

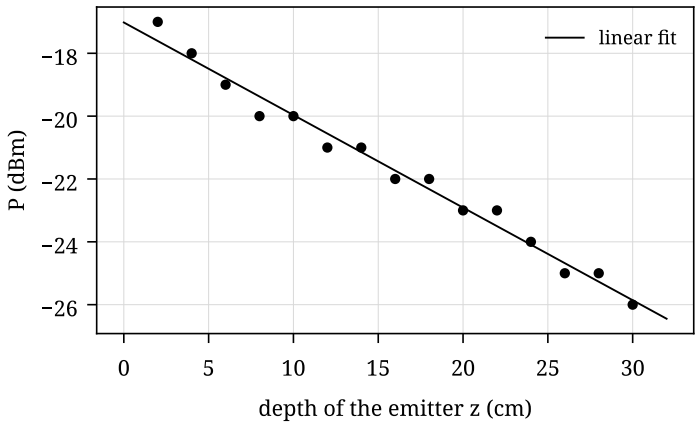


Figure S4. $P(z)$ in the water-filled tube.

Analysis. The fit gives a slope of $-29.5 \pm 1.1\,\mathrm{dB/m}$, hence

$$\alpha \approx \alpha_\mathrm{g} = 3.39 \pm 0.12\,\mathrm{m}^{-1}.$$

A finer point (not needed for full marks): in the guide the wave vector along the tube is $k = \sqrt{k_\mathrm{w}^2 - k_\star^2}$, a little less than $k_\mathrm{w} = 2\pi/\lambda_\mathrm{w}$, and the attenuation of the guided wave is $\alpha_\mathrm{g} = \alpha k_\mathrm{w}/k$. With $k_\star \approx 2 \cdot 1.84/d = 74\,\mathrm{m}^{-1}$ for $d = 50\,\mathrm{mm}$ and $k_\mathrm{w} = 166\,\mathrm{m}^{-1}$ from Task 2, $k/k_\mathrm{w} = 0.90$ and the attenuation of plane waves in water is $\alpha = 3.0\,\mathrm{m}^{-1}$. The few points nearest to the cap are left out, where the wave reflected by the bottom adds a ripple.

Task 4a. Decaying modes in air-filled waveguides. (2p)

Set-up. The metal tube lies on the table; the emitter is pushed in with the wooden rod Q (the rope pulls it back) to the distance z of its source line from the outlet, measured on the rod with the ruler; the receiver stands against the outlet.

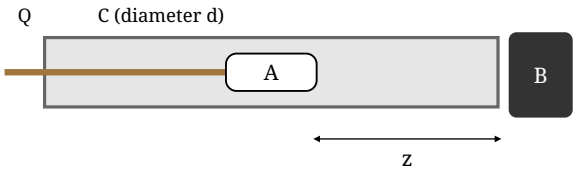


Figure S5. Set-up for Task 4.

Idea. Below cut-off $k = \mathrm{i}\mu$ and the field decays as $\mathrm{e}^{-\mu z}$, so $P_\mathrm{dBm} = \mathrm{const} - 8.686\,\mu z$. Close to the outlet ($z \lesssim 2\,\mathrm{cm}$) the faster decaying higher modes still contribute, and below about $-90\,\mathrm{dBm}$ the noise floor of Task 1 takes over; both ends are left out of the fit.

Table S3. P (dBm) against the distance z of the emitter from the outlet, for the four tubes.

z (cm)	$d_1 = 41$	$d_2 = 46$	$d_3 = 59$	$d_4 = 100$
1	−21	−21	−19	−17
2	−28	−26	−24	−19
3	−36	−33	−29	−22
4	−43	−40	−33	−24
5	−51	−46	−38	−27
6	−58	−53	−44	−30
7	−66	−60	−49	−32
8	−74	−67	−54	−35
9	−81	−73	−59	−37
10	−89	−80	−64	−40
11	−95	−87	−70	−43
12	−100	−93	−75	−46
13	−101	−99	−80	−49
14	−99	−99	−85	−52
15	−100	−100	−89	−54
16	−100	−100	−94	−57

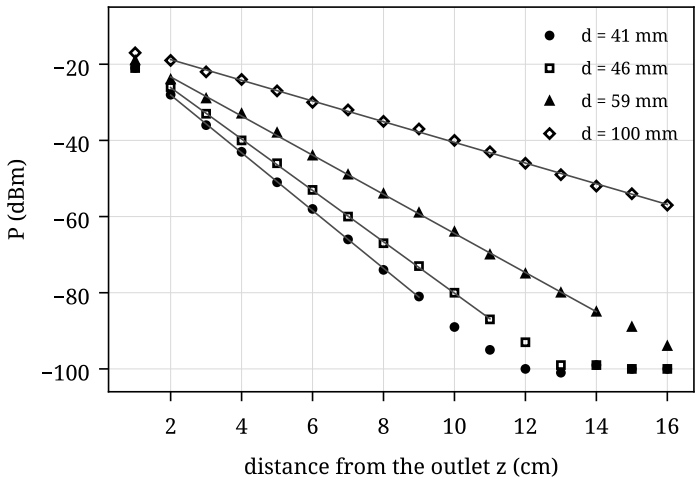


Figure S6. $P(z)$ in the four air-filled tubes; the lines are the fits.

Analysis. For $d = 46\,\mathrm{mm}$ the points between $z = 2$ and $12\,\mathrm{cm}$ lie on a line of slope $-6.75\,\mathrm{dB/cm}$:

$$\mu(46\,\mathrm{mm}) = 77.7 \pm 0.5\,\mathrm{m}^{-1}.$$

Task 4b. (5p)

The same measurement for all four tubes (Table S3, Figure S6) gives

	d (mm)	μ (m^{-1})	μd	μ^2 (m^{-2})	$1/d^2$ (m^{-2})
C1	41	87.3 ± 0.6	3.58	7622	595
C2	46	77.7 ± 0.5	3.57	6031	473
C3	59	59.1 ± 0.5	3.49	3498	287
C4	100	31.3 ± 0.5	3.13	979	100

Hypothesis. With $k = \mathrm{i}\mu$, the dispersion relation (2) reads $\mu^2 = k_\star^2 - \omega^2/c^2$. The only length the constant $k_\star$ can depend on is the diameter, so $k_\star = \xi/d$ with a number ξ:

$$\mu^2 = \frac{\xi^2}{d^2} - \left(\frac{2\pi}{\lambda_{\mathrm{air}}}\right)^2. \qquad (3)$$

The simpler guess $\mu \propto 1/d$ fails: μd is not constant but falls for the widest tube, where the second term matters. (3) predicts a straight line in the $(1/d^2, \mu^2)$ plane, and Figure S7 confirms it.

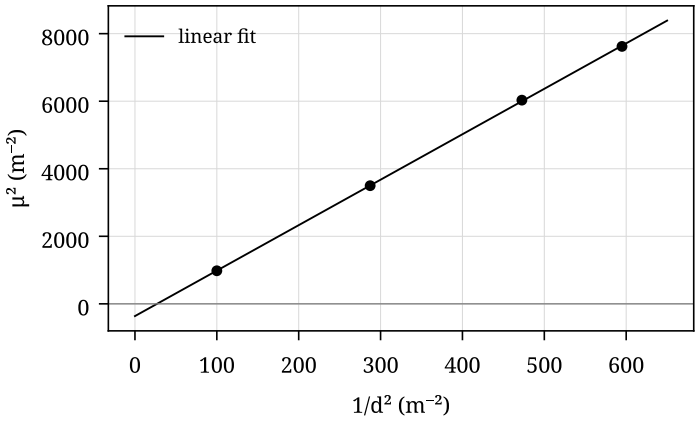


Figure S7. μ^2 against $1/d^2$.

The fit gives

$$\xi^2 = 13.46 \pm 0.08, \quad \xi = 3.669 \pm 0.010,$$

$$\text{intercept } -k_0^2 = -363 \pm 31\,\mathrm{m}^{-2}.$$

For comparison, the slowest-decaying mode of a circular metal tube, TE_{11}, has $\xi = 2 \cdot 1.841 = 3.682$.

Task 5. Wavelength in air and refractive index of water. (3p)

The intercept of the line in Figure S7 is $-(2\pi/\lambda_{\mathrm{air}})^2$:

$$k_0 = 19.0 \pm 0.8\,\mathrm{m}^{-1}, \qquad \lambda_{\mathrm{air}} = \frac{2\pi}{k_0} = 33.0 \pm 1.4\,\mathrm{cm}$$

(a frequency of $c/\lambda_{\mathrm{air}} \approx 0.91\,\mathrm{GHz}$). The refractive index of water for these waves is

$$n = \frac{\lambda_{\mathrm{air}}}{\lambda_{\mathrm{w}}} = 8.7 \pm 0.4,$$

close to $\sqrt{\varepsilon} \approx \sqrt{80} \approx 8.9$ for water at radio frequencies. The wavelength in air can also be measured directly: with a flat sheet of foil as a mirror in front of the emitter, the reading as the receiver moves between them has minima $\lambda_{\mathrm{air}}/2 \approx 17\,\mathrm{cm}$ apart, which agrees with the value above.